MHD-WAAM THESIS | EXACA DATA NOTE

The E1 Dataset and ExaCA

What the first ExaCA-ready run can and cannot be used for, explained from the basics up, and which files go to the ExaCA side

20 September 2026 | run waam-arc-01-E1-exaca (model state M2, 0.5 mm mesh, 2.0 s of welding) | for the thesis team

THE SHORT VERSION

1. E1 is the first simulation of the thesis that writes the data ExaCA needs. The file is clean: 3,610 melting-and-freezing records in 1,231 cells, no damaged lines, every record in the right time order.

2. But it covers only **19% of the bead**. The cells that have records form a column about 2 mm wide down the middle, where the droplets land, plus the melted plate underneath. ExaCA treats every cell missing from the file as metal that never melted, so it would grow new grains only in that column and leave the other 81% of the bead untouched.

3. So E1 is for **setting up and testing** the ExaCA side (reading the file, the grid, the domain, the Al-5Mg material file, the settings and the analysis), **not for grain-structure results**.

4. Found while writing this note: current ExaCA (version 1.3 and later) needs the temperature data on its own fine grid. Ours is on a 0.5 mm grid, so it has to be **resampled** before any run that resolves grains. As it is, it can only be read with 500 micron ExaCA cells, which is a useful first test and nothing more.

5. The files to send are listed in Section 10.

How to read this. Sections 1 to 3 are background: what the thesis simulates, how metal freezes, and what ExaCA does. Sections 4 to 7 are about E1: how its data was made, what the checks found, why it covers so little of the bead, and what that means for ExaCA. Sections 8 to 10 are practical: what E1 is good for, what happens next, and what to send. Every technical word is explained where it first appears and again in the glossary (Section 11).

1. The big picture

1.1 What the thesis simulates

Wire arc additive manufacturing (WAAM) builds metal parts a little like a hot-glue gun builds a line of glue, except that the heat comes from an electric arc and the material is a metal wire. The arc melts the tip of the wire. The molten tip breaks off as droplets, about 30 of them per second in the process we copy, and each droplet falls into a small pool of molten metal on the plate below. The welding torch moves along at 20 mm per second. Behind it the pool freezes into a solid ridge of metal called a **bead**. Laying beads side by side and on top of each other builds up a part.

The thesis copies one published experiment, Zhao et al. (2021): a single bead of Al-5Mg (aluminium with about 5% magnesium) laid on a 4 mm thick plate, at 135 A and 16 V, with a wire 1.2 mm in diameter fed at 150 mm/s. Our simulation runs in OpenFOAM, using a modified version of the solver laserbeamFoam. It calculates how the molten metal and the surrounding argon gas flow and how hot every point is. To do that,

space is divided into small cubes called **cells**, 0.5 mm on a side, and time is advanced in steps of 10 microseconds (0.00001 s). Once the model matches the experiment, the thesis adds a magnetic field and asks how it changes the pool and the metal the pool leaves behind.

1.2 Why a second program is needed

What we care about in the end is the **microstructure**: the pattern of tiny crystals, called **grains**, that forms when the metal freezes, and the finer structure inside each grain, which grows as tree-like branches called **dendrites**. It decides much of the strength and toughness of the part, and it is what a magnetic field is expected to change. Zhao et al. (2022) deposited the same alloy with and without an alternating magnetic field. With the field, the bottom of their walls had more rounded (equiaxed) grains and a finer structure: the largest spacing between dendrite arms fell from about 13 micron to about 9 micron (1 micron is a thousandth of a millimetre).

Grains are tiny compared with our cells. One 0.5 mm cell is 500 micron across, so it could contain dozens of grains, and the flow simulation cannot see them at all. To see grains you need cells of about 5 to 10 micron: 50 to 100 times finer in each direction, which means 125,000 to 1,000,000 times as many cells. Running the flow simulation at that resolution is out of reach of any computer available to us. So the work is split between two programs:

- the **flow simulation** (our side, OpenFOAM) works out when every part of the metal melts, how long it stays liquid and how fast it cools, on the coarse 0.5 mm grid;
- **ExaCA** (the thesis partner's side) takes that temperature history and grows the grains on a very fine grid. It can afford the fine grid because it does not calculate flow or heat at all. It only applies simple rules for how crystals grow, which is far cheaper per cell.

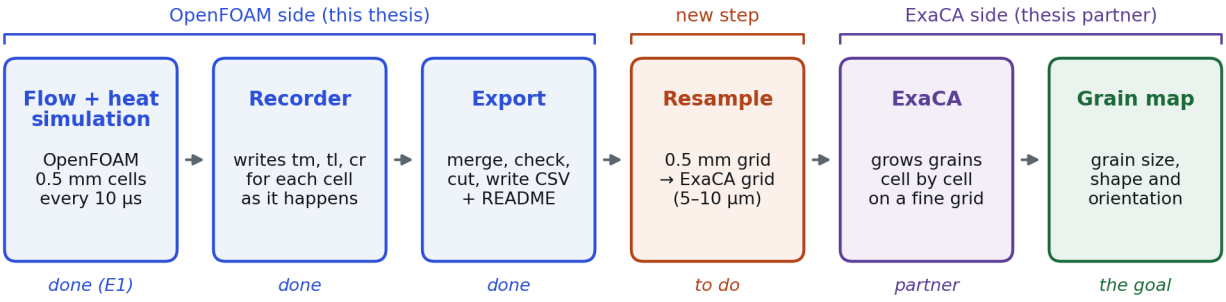


Figure 1. The chain from the welding simulation to a grain map. The blue steps exist and produced E1. The resampling step (orange) turned out to be necessary while this note was being written (Section 3.3). Purple and green are the ExaCA side.

2. Background: how metal freezes

2.1 An alloy freezes over a range of temperatures

Pure water freezes at a single temperature, 0 °C, and so does pure aluminium, at 660 °C. An alloy such as Al-5Mg does not. It starts to freeze at one temperature and finishes at a lower one. Temperatures in this note are in kelvin (K), the scientific scale: 0 °C is 273 K, and a change of 1 K is the same as a change of 1 °C.

name	value	what it means
liquidus	906 K (633 °C)	above this the metal is completely liquid
solidus	815 K (542 °C)	below this it is completely solid
mushy zone	the 91 K band between them	a mixture of solid crystals and liquid, like slush; the lower the temperature, the more of it is solid

The reason for the range is that the first crystals to form take in less magnesium than the liquid holds. The magnesium they push out stays in the remaining liquid, which becomes richer in magnesium, and magnesium-rich liquid freezes at a lower temperature. So freezing continues downwards through the range as the liquid becomes richer, until the solidus. (The values above are the ones our simulation uses, taken from Zhao et al. 2021.) The mushy zone matters a great deal in this note: the recorder that produced E1 only starts timing a cell when it is above the liquidus, and most of our simulated pool turns out to be mushy.

2.2 What a grain is

When liquid metal freezes, its atoms line up into a regular three-dimensional pattern, a crystal lattice. They do not line up into one single crystal for the whole bead. Crystals start growing at many places at once, each with its lattice pointing in its own direction, and they keep growing until they run into each other. Each of these crystals is a **grain**, and the surfaces where two grains meet are grain boundaries. A polished and etched cross-section of a bead looks under the microscope like a mosaic of irregular tiles.

Three features of the grains matter for the finished part. Their **size**: smaller grains usually make a stronger and tougher metal. Their **shape**: long thin *columnar* grains, or rounded *equiaxed* grains that are about the same size in every direction. And their **orientation**: if most lattices point the same way (called *texture*), the part is stronger in some directions than in others.

2.3 Where grains start, and which way they grow

At the edge of the pool, the metal touching the liquid is solid metal that did not melt: the plate, or an earlier layer. It already contains grains. The freezing liquid simply continues those grains, copying their lattice direction. This is called **epitaxial growth**. Grains grow towards the hottest part of the pool, against the direction in which heat is flowing out. Crystals of aluminium grow fastest along the edges of their cubic lattice, so grains whose fast direction happens to point along the heat flow outrun their neighbours and crowd them out. That competition is why welds and WAAM beads often contain long columnar grains pointing towards the middle of the pool.

New grains can also appear in the liquid itself, ahead of the growing ones, when the liquid is cold enough and contains something for a crystal to start on, such as a tiny particle or a broken-off crystal tip. This is **nucleation**. A lot of nucleation gives many small equiaxed grains; very little gives long columnar grains.

2.4 Undercooling and the interfacial response function

A freezing front cannot advance at exactly the liquidus temperature. The liquid just in front of it has to be slightly colder than the liquidus, and the amount by which it is colder is called the **undercooling**. The larger the undercooling, the faster the crystal tips grow. The relation between undercooling and growth speed is a property of the alloy, described by a curve called the **interfacial response function**. ExaCA needs this curve for Al-5Mg, and it is not supplied with ExaCA, so it has to be worked out (Section 8.1).

2.5 The gradient G, the growth rate R, and the cooling rate

Two numbers at the freezing front control the shape and size of the grains:

- **G, the temperature gradient**: how quickly the temperature rises with distance into the liquid at the front, in kelvin per millimetre (K/mm) or per metre (K/m). A large G means the liquid gets much hotter within a short distance.
- **R, the growth rate**: how fast the freezing front moves, in mm/s or m/s.

Their product is the **cooling rate** felt at a fixed point as the front passes:

```
cooling rate (K/s) = G (K/mm) × R (mm/s)
```

To see why, imagine standing still inside the metal while the freezing front sweeps past you at speed R. The pattern of temperatures travels with the front. If the temperature falls by G kelvin for every millimetre, and R millimetres of that pattern pass you each second, then your temperature falls by G × R kelvin every second. With numbers typical of E1: G = 78 K/mm and R = 27 mm/s give 78 × 27, about 2,100 K/s. The rules of thumb from solidification science are: a large ratio G/R favours columnar grains and a small one favours equiaxed grains, and faster cooling (a larger G × R) gives a finer structure.

3. Background: what ExaCA does

3.1 A cellular automaton, in plain words

ExaCA (short for Exascale Cellular Automata) is a free, open-source program written at the US national laboratories, first at Oak Ridge and now maintained at Lawrence Livermore (github.com/LLNL/ExaCA), to predict the grain structure of metal made by additive manufacturing. A **cellular automaton** is a simple kind of simulation: space is divided into a grid of identical small cubes, each cube is in one of a few possible states, and at every time step each cube updates its state by a few rules that look only at its neighbours.

In ExaCA each cube is typically 2.5 to 10 micron on a side, and it is either liquid, or solid and belonging to a numbered grain with that grain's lattice orientation, or on the growing edge of a grain. The rules are, in words:

1. **Melting.** At the time the temperature data says the location melted (tm), the cube becomes liquid, ready to be taken over by whichever grain reaches it first when it freezes again.
2. **Cooling.** From the time the location cools back through the liquidus (tl), its undercooling grows steadily at the cooling rate cr.
3. **Growth.** A grain on the edge of the liquid grows into its liquid neighbours at the speed that the interfacial response function gives for the current undercooling. The growing shape is an octahedron (a double pyramid) lined up with the grain's own lattice, so each grain grows fastest along its own preferred directions. This is how the competition between grains described in Section 2.3 comes out of the simulation.
4. **Nucleation.** Scattered through the liquid are possible starting points for new grains, each with its own trigger undercooling (drawn at random around a chosen average). When the local undercooling reaches a site's trigger, a new grain with a random orientation appears there.

3.2 What ExaCA needs from the flow simulation

ExaCA does not calculate heat flow. For each location it is given three numbers per melting:

name	meaning	unit
tm	the time the location went above the liquidus (melted)	s
tl	the time it went back below the liquidus (started to freeze)	s
cr	the cooling rate at the liquidus, at time tl	K/s

together with the position of the location, x, y and z, in metres. If a location melts more than once, it appears once for every melting, on separate lines. This is ExaCA's standard input for the problem type it calls *FromFile*: a plain text file (a CSV file, meaning comma-separated values) whose first line is `x,y,z,tm,tl,cr` and whose every other line holds six numbers separated by commas.

From tl and cr, ExaCA rebuilds the temperature after freezing starts with a straight-line assumption:

```
undercooling at time t = cr × (t - tl) for t later than tl
```

In words: at the moment tl the location is exactly at the liquidus, with zero undercooling, and every second after that it is cr kelvin colder. With cr = 2,000 K/s it is 2 K below the liquidus one millisecond after tl, and 10 K below after five milliseconds. The straight line is an approximation, because real cooling slows down as the metal releases its latent heat and as it approaches the plate temperature. It is good for the first few kelvin of undercooling, which is where the grain structure is decided, and that is why cr is measured right at the liquidus.

3.3 Two consequences that decide everything in this note

(a) A location missing from the file never melts in ExaCA. At the start, ExaCA fills its whole box with solid metal: the region below the plate top gets a starting grain pattern of a chosen size (ExaCA calls it the baseplate), and the region above it gets a 'powder' pattern (by default one tiny grain per ExaCA cube). Only the locations listed in the file are melted and re-grown. Everything else keeps its starting grains for the whole run, and ExaCA marks it as never melted (its LayerID output is -1). For the plate that is correct: plate metal that never got above the liquidus really does keep its grains. For the gas above the bead it does no harm, provided the analysis only looks at cells that melted. But for bead metal missing from the file it is wrong, because every bit of the bead was liquid when it left the wire.

(b) ExaCA needs the data on its own grid. Since version 1.3 (February 2024), and in the current 2.x releases, ExaCA reads the temperature data at exactly the size of its own cubes. The older input for a coarser thermal grid, HeatTransferCellSize, is now ignored with the message 'temperature data must be at the same resolution as the CA cell size'. Even in the older versions it could not be combined with re-melting, which our data contains. Our data is on a 0.5 mm grid. If ExaCA were given it with 10 micron cubes, each of our points would land in one single cube, 50 cubes away from the next point in every direction, and the other 124,999 cubes in each of our cells would get no data, so they would never melt (Figure 2b). The data therefore has to be **resampled** onto the fine grid first (Figure 2c). The only ExaCA setting that can read E1 as it is, is a cube size of 500 micron: that tests the file reading, but it is far too coarse to show a single grain. (The README of the first export says HeatTransferCellSize = 500 micron. That describes our grid correctly, but current ExaCA does not use it.)

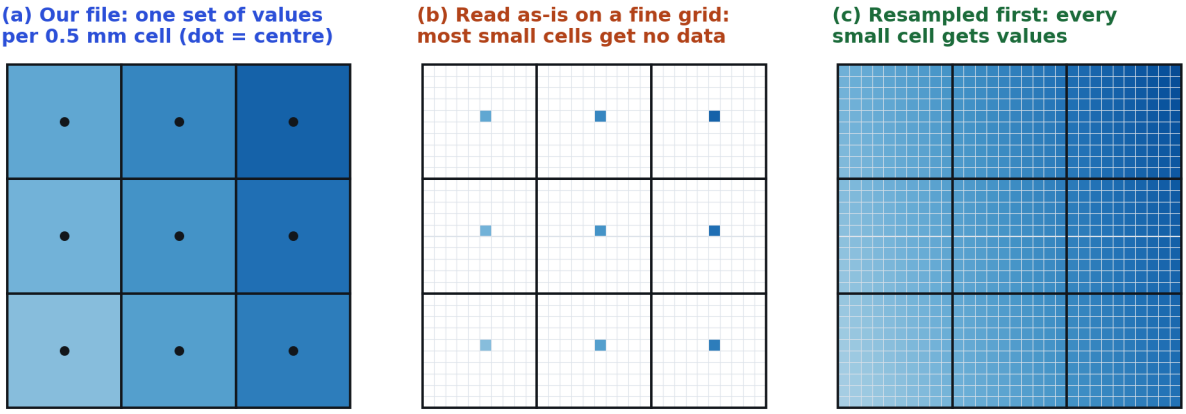


Figure 2. Why resampling is needed. (a) Our file has one set of values per 0.5 mm cell. (b) Given directly to ExaCA with fine cubes, each point fills a single cube and the rest of the cell gets nothing. (c) Resampling gives every fine cube its own values. Drawn with 10 × 10 fine cubes per cell so that it stays readable; at 10 micron the real ratio is 50 per side, 125,000 cubes per cell.

4. How E1 produced its data

4.1 The run

item	E1
case name	waam-arc-01-E1-exaca
physics	model state M2, unchanged: the arc puts 1,113.7 W into the pool, the droplets bring another 722 W, and the electromagnetic (Lorentz) force of the welding current is on
mesh	98 × 50 × 18 = 88,200 cells. Uniform 0.5 mm cubes in the core box x = -12 to 32 mm, y = -6 to 6 mm and the full height z = -4 to 5 mm; stretched cells only outside it
coordinates	z = 0 is the top of the 4 mm plate and +z points up; y = 0 is the bead centreline; the torch starts at x = -10 mm at t = 0 and moves in +x at 20 mm/s
length	2.0 s of welding, which is Zhao's full 40 mm track: 200,000 time steps of 10 microseconds
cost	38,038 s, i.e. 10.6 hours, on 6 processor cores of the local PC
difference from M2	only two: the end time, and the recorder of Section 4.2 switched on

M2 is the model state that best matches the experiment so far. Its liquid pool reaches about 0.9 mm into the plate, close to Zhao's measured 1 mm, and the underside of the plate stays solid, as in the experiment. Its known weakness is width: the zone of plate that melts is about 3.6 mm wide, where Zhao measured about 7.8 mm. That weakness comes back in Section 6.

4.2 The recorder

The recorder is a short piece of C++ code (an OpenFOAM 'coded function object') that is compiled when the run starts and runs after every time step. It does not change the physics; it only watches. A useful picture is a clerk standing at each of the 88,200 cells with a thermometer and a stopwatch, looking every 10 microseconds and writing down what happens.

The clerk can read two numbers in each cell: the temperature, and how much of the cell is metal. The second one needs a word of explanation. The simulation keeps track of where the metal is with the **volume of fluid** method: every cell stores the fraction of its volume that is filled with metal, from 0 (only argon) to 1 (only

metal). Cells at the surface of the metal are partly filled. The recorder counts a cell as metal when this fraction, called alpha.metal in the code, is more than one half. The rules are:

	situation	what the recorder does
1	The cell is metal and at or above 906 K, and no record is open.	Opens a record and writes down **tm**: the moment the cell crossed 906 K, or the moment the metal arrived if it arrived already hotter than that (a fresh droplet).
2	A record is open and the temperature falls below 906 K.	Writes down **tl**, the exact crossing time (found by interpolating between the two time steps either side), and **G**, the temperature gradient at that moment.
3	A record is open and the temperature reaches 896 K.	Works out **cr** = 10 K ÷ (time taken to cool from 906 K to 896 K). The record is complete and is written as one line of the output.
4	The temperature goes back above 906 K before it reached 896 K.	The dip did not count as freezing. It keeps waiting, as part of the same melting.
5	A record is open and the metal fraction falls below one half.	Throws the record away. It cannot tell whether the metal has flowed out of the cell or the surface has only rippled through it.
6	A completed cell melts again later.	Starts a new record. One cell can have several.

Why 10 K. The cooling rate is the slope of the temperature against time just below the liquidus. Measuring it across 10 K, rather than between two neighbouring time steps, averages away small wiggles, and it matches what ExaCA uses the number for: the first few kelvin of undercooling. It also filters out false freezing: a cell that dips a few kelvin below the liquidus and is re-melted by the next droplet has not really solidified (rule 4).

Why every time step. The recorder looks 100,000 times per simulated second instead of using the snapshots the simulation saves every 0.02 s. The droplets arrive 30 times a second, and anything that repeats 30 times a second needs to be looked at more than 60 times a second to be followed at all. Snapshots at 50 per second would blur successive droplets together.

Each of the 6 processor cores writes its own file. When the run ended there were 3,610 completed records, 68 cells still had a record open (still hot or still cooling), and 8,064 records had been thrown away under rule 5.

4.3 From recorder files to the package: the export

A Python script, export_exaca.py, turns the recorder's files into the package for ExaCA. In order, it:

1. merges the six files, one per processor core;
2. drops records whose melting time is unknown (this can only happen when a run is restarted part-way; there were none);
3. keeps only cells inside the uniform 0.5 mm core of the mesh, and checks that every cell centre sits exactly on the regular 0.5 mm lattice (centres at 0.25 mm, 0.75 mm, 1.25 mm and so on), because ExaCA needs a regular grid;
4. optionally cuts out a slab between two x positions;
5. sorts the records by position and then by time, so the meltings of one cell are consecutive and in order;
6. writes three files: the ExaCA temperature file (x,y,z,tm,tl,cr), a companion file with G and R for every record (for plots; ExaCA does not read it), and a README with all the definitions.

5. What the checks found: the good news

check	result	what it means
damaged lines in the recorder files	0	all six cores wrote their files cleanly
records off the regular grid	0	every record sits on the 0.5 mm lattice
records that froze before they melted (tl earlier than tm)	0	the time order is right in every record
cells with records	1,231, of which 916 melted more than once	2.9 records per cell on average; explained below
melting times tm / freezing times tl	0.067 to 1.978 s / 0.072 to 1.991 s	the records cover the whole 2 s run
cooling rate cr	364 to 33,280 K/s, median 2,099 K/s	typical of arc deposition
gradient G	1.2 to 495 K/mm, median 78 K/mm	see the warning in Section 7
plate underside, 1.8 to 2.0 s	757 to 758 K, flat	57 K below the solidus: the plate does not melt through, as in the experiment
depth of the liquid pool	0.75 to 1.25 mm, alternating with the droplets	Zhao: about 1 mm

Why most cells melted about three times. A droplet lands every 1/30 of a second, which is 33 milliseconds. In that time the torch moves 20 mm/s × 0.033 s = 0.67 mm. Each droplet, however, is about 2.2 mm across. So every spot on the centreline is hit by roughly three droplets in a row (2.2 ÷ 0.67 = 3.3), and each one re-melts it. The same happens in real deposition, and it is one source of the fine layered bands seen in WAAM microstructures. ExaCA handles it: a location that melts several times simply appears on several lines.

5.1 Where along the track the bead is steady

The start of a weld is not typical: the plate is cold, so the first metal is quenched hard. Only after some distance do the conditions settle. Figure 3 shows the records along the track in 2 mm bins: how many there are, and the median (middle) cooling rate and gradient in each bin.

x (mm)	what is happening
-12 to -4	**start**: the cold plate quenches the first metal; cooling rates 2,700 to 17,000 K/s
-4 to 6	**settling**: the cooling rate has settled near 2,000 K/s, but G is still climbing, from 52 towards 80 K/mm
6 to 26	**steady**, 20 mm of track: cooling rate 1,830 to 2,120 K/s, G 75 to 85 K/mm
26 to 30	**run ends**: the simulation stopped at 2.0 s while this metal was still hot, so many of its records were never completed

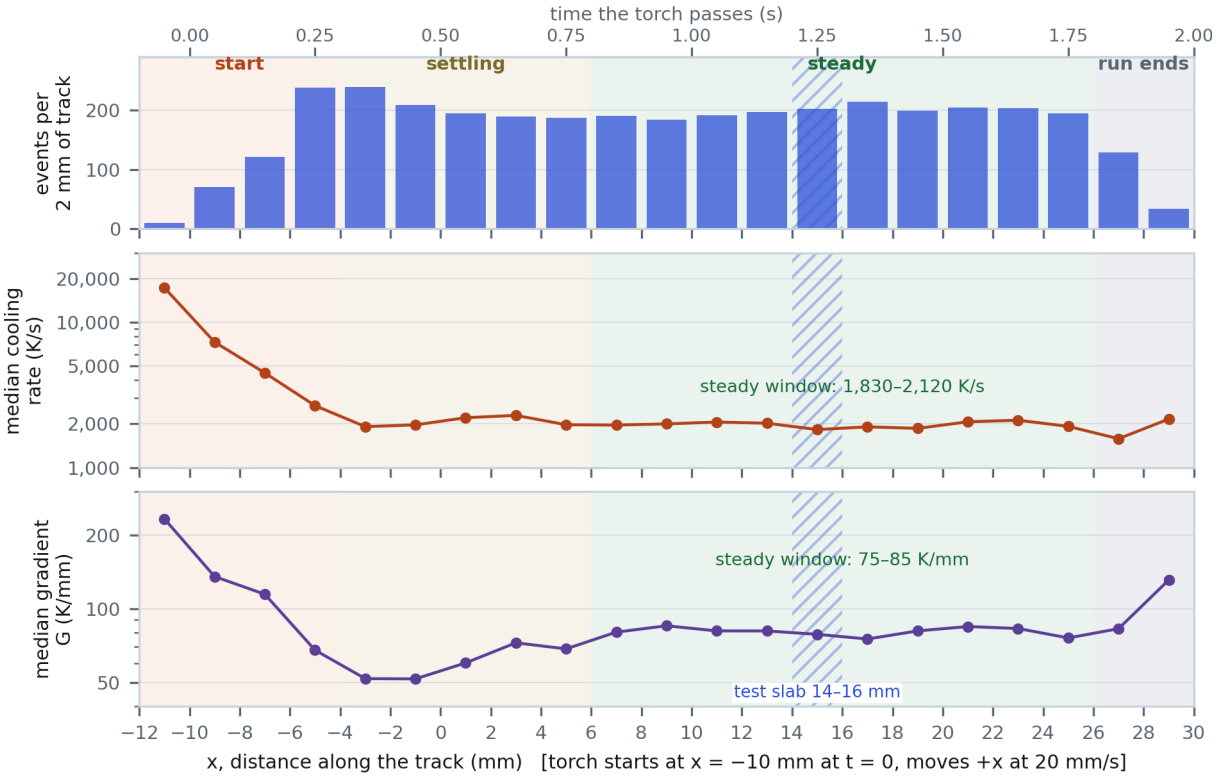


Figure 3. E1 records along the track, in 2 mm bins (real data from the export). Top: number of records. Middle: median cooling rate. Bottom: median gradient G. The top axis gives the time at which the torch passes each position, t = (x + 10 mm) ÷ 20 mm/s. Hatched: the suggested test slab, x = 14 to 16 mm.

'Steady' means that the conditions for freezing no longer change along the track. The plate as a whole keeps warming slowly (its far edge rose from 466 K to 480 K over the last 0.2 s), which is expected for a small plate. The suggested test slab is x = 14 to 16 mm, in the middle of the steady window and 8 mm from either end. The torch passes x = 15 mm at t = 1.25 s.

6. The problem: the records cover 19% of the bead

6.1 How coverage was measured

A clean file is not enough. It also has to describe all the metal that froze. To check, we took the metal as it is at the end of the run (t = 2.0 s), in the steady window x = 6 to 26 mm (41 slices of 0.5 mm), and asked of every bead cell above the plate (z above 0) whether it has at least one record. All of the bead was delivered as liquid droplets, so every one of these cells was liquid at some point, and ideally every one should have a record.

quantity, steady window x = 6 to 26 mm	count	per 0.5 mm slice
bead cells above the plate (metal at t = 2.0 s)	1,234	30
...of which have a record	234 (19%)	6
...of which have no record	1,000 (81%)	24
plate cells below z = 0 with a record (the melted plate)	406	10

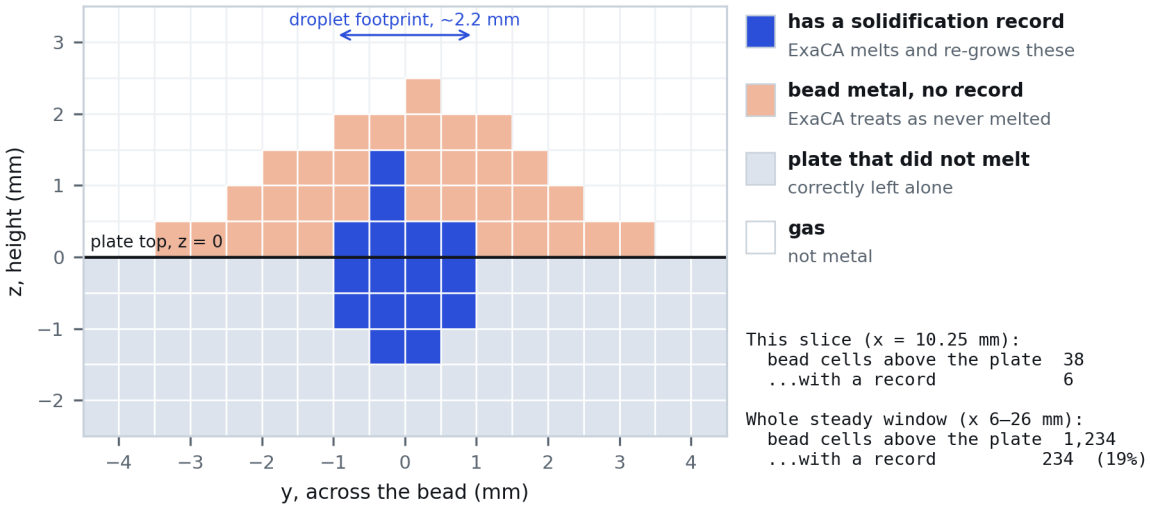


Figure 4. Cross-section of the bead at x = 10.25 mm, at the end of the run (real data). Each square is one 0.5 mm cell. Blue cells have at least one record; orange cells are bead metal with none; grey is plate that never went above the liquidus; white is gas.

The cells with records form a column about 2 mm wide at the base of the bead, narrowing to a single 0.5 mm cell higher up, plus a region about 2 mm wide and 1.5 mm deep in the plate underneath. That is the footprint of the droplets, which are about 2.2 mm across and land on the centreline. The rest of the bead, the sides and most of the upper part, has no record.

6.2 Why: three kinds of cell history

The recorder opens a record only when a cell is metal *and* above 906 K at the same moment. Figure 5 shows the three kinds of history that occur in E1.

(a) Under the droplets. Droplets arrive at about 1,500 K, far above the liquidus. The cell they land in is filled with liquid metal, stays liquid for a while under the arc, and then cools through 906 K and 896 K where it is. Every rule of the recorder is satisfied and the record is completed. This is the blue column in Figure 4.

(b) Metal that arrives already mushy. The sides of the bead are built by metal that flows out sideways from where the droplets land. On the way it loses heat to the much cooler plate and mixes with cooler metal. In E1 it has usually dropped below 906 K by the time a side cell is more than half full. From the recorder's point of view, that cell went straight from 'gas' to 'mushy metal'. It never saw the cell above the liquidus, so it never opened a record. The metal did melt, but somewhere else, a few milliseconds earlier.

(c) Records thrown away at the surface. The surface of the pool ripples, and every droplet impact sends waves across it. A cell at the surface can be more than half metal one moment and less than half the next. Rule 5 then throws its open record away, because the recorder cannot tell a ripple from metal really flowing out of the cell. This happened 8,064 times in E1 (the count is of records, not of cells: one cell can lose many). When the metal comes back, it is usually below 906 K by then, so no new record opens.

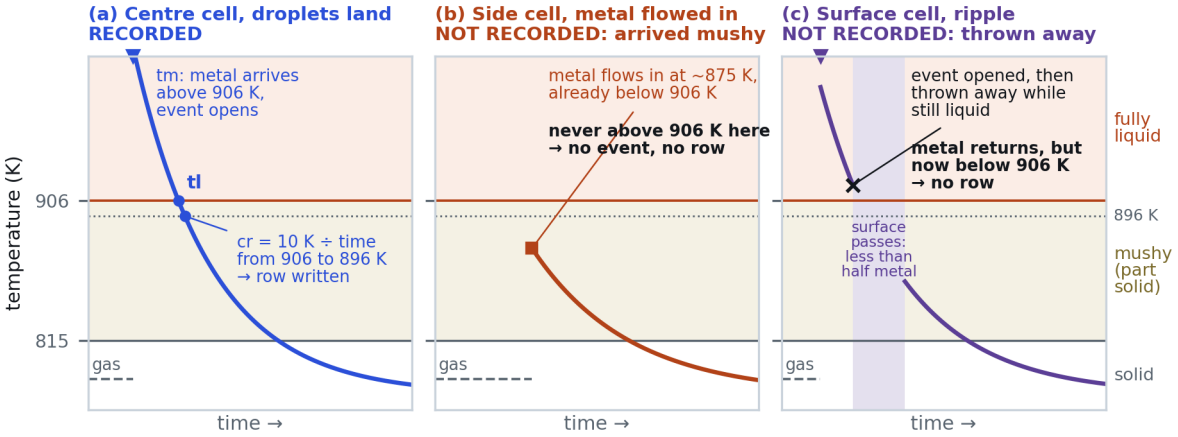


Figure 5. Sketches of three cell histories (shapes are illustrative, not simulation output). (a) Under the droplets, metal arrives far above the liquidus and cools through it in place, so a record is opened and completed. (b) At the sides, metal flows in already below the liquidus, so no record opens. (c) At the rippling surface, a record opens and is thrown away when the cell drops below half metal; when the metal returns it is too cool to open a new one.

We do not yet know how much of the missing 81% each of the two causes, (b) and (c), accounts for. The next version of the recorder counts them separately (Section 9).

6.3 The deeper reason: a mostly mushy pool

Both causes are made worse by a feature of the M2 model: its pool is mostly mush. At the end of the run the pool, meaning all the metal above the solidus (815 K), filled 436 cells. The recorder had only 68 records open at that moment, and every metal cell above 906 K has an open record, so at most 68 of the 436 pool cells were fully liquid: at least five cells in six were mushy. In the experiment the melted zone is about 7.8 mm wide, twice the width of M2's, which means liquid metal reached much further out to the sides. This is the same shortfall of heat at the sides that makes M2's melted zone too narrow, seen from a new angle. It also makes coverage a useful new check on the model: a pool that is right should give close to 100%.

6.4 So is the recorder wrong?

Its arithmetic is right; its definition is too narrow for this situation. 'A cell melts when it goes above the liquidus' is exactly ExaCA's own definition, and it is the right one for the plate: a plate cell that only became mushy keeps most of its original solid, and ExaCA correctly leaves it alone. It is the wrong rule for the bead, because bead metal did not melt in the cell where it ends up. It arrived there as liquid from the wire, usually partly frozen already, and finished freezing in place. The fix (Section 9) is to record the arrival of mushy metal as the start of a freezing event too.

7. What this means for ExaCA

Suppose E1 were resampled onto a fine grid and given to ExaCA as it is. Then:

- in the centre column and the melted plate beneath it, ExaCA would melt the metal and grow new grains, starting from the plate's grains at the bottom of the melted region, as it should;
- in the other 81% of the bead nothing would happen. Those cells would keep the starting pattern ExaCA gave them (above z = 0 that is the 'powder' pattern, by default one tiny grain per cube), and ExaCA would mark them as never melted;

- a picture of the result would show a narrow strip of real solidification grains down the middle of a bead whose sides are either speckle or blank. Any grain statistics, whether size, shape or texture, would describe the centre column only, which is also the part of the bead most disturbed by the droplets.

There is a second, smaller warning about the cells that do have records. The speed of the freezing front can be worked out from each record as R = cr ÷ G (Section 2.5 turned around). In a steady bead the freezing front at the back of the pool travels along with the torch, so it can never move faster than the torch, 20 mm/s. At the very back of the pool it moves at exactly the travel speed. Along the sides and the bottom the front is slanted, and a slanted front moves more slowly measured square to itself:

```
R = travel speed × cos(theta) at most 20 mm/s
```

where theta is the angle between the direction the front faces and the direction of travel (cos is 1 when theta is 0 and falls towards 0 as the front turns side-on). The median R in E1 is 33 mm/s, faster than the torch. So many of the records do not come from the steady tail of the pool. They come mostly from freshly landed droplet metal freezing quickly after re-melting the metal beneath. On top of that, a 0.5 mm cell smears a steep temperature front over about two cells, which makes G come out too small and therefore R too large. The 0.25 mm runs will sharpen G.

> **BOTTOM LINE FOR THE ExaCA SIDE**
>
> E1 cannot answer any question about grain size, shape or texture. Any grain map made from it is a test of the machinery, not a result, and should not appear in the thesis as one.

8. What E1 is good for

Almost everything needed on the ExaCA side does not depend on how good the temperature data is, and every later dataset will come in exactly the same format. So E1 is the right dataset to set everything up with now, while the better data is being made.

task	what to do with E1 now	carries over?
Install and build ExaCA	ExaCA runs on ordinary processors or NVIDIA graphics cards through the Kokkos library. Build it once; the E1 test runs are small.	yes
Read the file	Run the full-track file with CellSize = 500 micron. This checks the header, the units (metres, seconds, K/s), the re-melting lines, and that the domain comes out as expected: cell centres from x = -11.25 to 29.75 mm, y = -1.75 to 2.25 mm and z = -1.25 to 2.75 mm.	yes
Coordinates	Plate top at z = 0, which is ExaCA's default BaseplateTopZ, so there is nothing to set; +z up; y = 0 on the centreline; x along the direction of travel.	yes
Choose the domain	A box around the whole track would be 844 million cubes at 10 micron (6.7 billion at 5 micron), far too many for a desktop. Use a 2 mm slab from the steady window. E1's slab, x = 14 to 16 mm, spans about 2 × 4.5 × 4.5 mm: roughly 40 million cubes at 10 micron (the export prints the exact number). ExaCA's XRegion and YRegion inputs can also cut a region out of the full file. The calibrated bead will be about twice as wide, so plan for about 70 million.	yes; sizes grow
Resample	Try the resampling step on the slab as soon as the tool exists (Section 9).	yes
Al-5Mg material file	Derive the interfacial response function (Section 8.1). Freezing range 91 K, the same as the flow simulation.	yes; the biggest single job

task	what to do with E1 now	carries over?
Nucleation and starting grains	Choose the density of nucleation sites and the average and spread of their trigger undercooling, and the grain size of the plate (5A05 alloy) for the baseplate pattern. These have to be calibrated later against measured grain structures. Zhao et al. (2022) report dendrite arm spacings, not grain sizes, so a grain-size target still has to be taken from their micrographs or another Al-5Mg WAAM study. (The 80 to 31 micron quoted in some earlier project notes is their largest pore diameter, not a grain size.)	yes
Time step, run time, memory	Measure them on the slab, to plan the real runs.	yes
Output and analysis	Maps of grain size, shape (length to width, the aspect ratio) and orientation. Restrict every statistic to cells that melted (LayerID not equal to -1).	yes

8.1 The material file, in a little more detail

ExaCA's material file is a short text file in JSON format. The example shipped with ExaCA for the nickel alloy Inconel 625 reads:

```
{
   "function": "cubic",
   "coefficients": { "A": -0.00000010302, "B": 0.00010533,
                     "C": 0.0022196,      "D": 0 },
   "freezing_range": 210
}
```

The Inconel 625 material file shipped with ExaCA (examples/Materials/Inconel625.json).

It says that the speed V of a growing crystal tip, in m/s, depends on the undercooling dT, in K, through a cubic polynomial, and that the alloy freezes over a 210 K range:

$$V = A \times dT^3 + B \times dT^2 + C \times dT + D$$

A cubic is simply a smooth curve with four adjustable numbers, A to D, fitted through values calculated from theory. For Inconel 625, an undercooling of 10 K gives V = 0.033 m/s, i.e. 33 mm/s. ExaCA also accepts quadratic, power-law and exponential curves. For Al-5Mg no such curve is supplied, so it has to be derived. The standard route is the dendrite-growth theory of Kurz, Giovanola and Trivedi (the KGT model, 1986). It needs four properties of the alloy:

- the **liquidus slope**: how much the liquidus temperature drops per percent of magnesium, read from the aluminium-magnesium phase diagram;
- the **partition coefficient**: how much less magnesium the first solid takes in than the liquid holds (Section 2.1);
- the **diffusion coefficient** of magnesium in liquid aluminium: how fast the rejected magnesium spreads away from the growing tip;
- the **Gibbs-Thomson coefficient**: how much a sharply curved crystal tip lowers its own melting point.

KGT gives V for a range of undercoolings, and the cubic is then fitted through those points. The freezing_range entry should be 91 K (906 K minus 815 K), so that ExaCA and the flow simulation describe the same alloy.

The current development version of ExaCA (not yet in release 2.1) also ships a file for an aluminium alloy, AlSi10Mg: a power law, $V = A \times dT^{B}$ with $A = 7.3 \times 10^{-8}$ and $B = 3.34$, and a freezing range of 31 K. It is a different alloy (silicon instead of magnesium, and a freezing range a third as wide), so it must not be used for results. But it can stand in during the first test runs, so that the file reading, the domain and the output can be checked while the Al-5Mg file is being derived.

9. What happens next

9.1 A resampling tool (new)

A script that turns the 0.5 mm records into records on the ExaCA grid for a chosen slab. It should write ExaCA's binary format (a file ending in .catemp: the same six numbers per line, stored as raw double-precision values instead of text, which is much smaller and faster to read). There are two sensible ways to fill the fine grid:

- **copy**: every fine cube takes the records of the 0.5 mm cell it lies in. It is simple and keeps the re-melting, but it produces 0.5 mm blocks that start freezing all at the same instant, which is fine for testing the chain and wrong for results;
- **smooth**: tm, tl and cr are interpolated between neighbouring cell centres (a weighted average of the eight surrounding centres, weighted by closeness, called trilinear interpolation), so that the freezing front moves smoothly through the fine grid. Near the edge of the melted region this needs one more number from the recorder, each record's peak temperature, so that the edge can be placed where the interpolated peak temperature equals 906 K. Re-melting complicates it, because neighbouring cells can have different numbers of records; the simplest sound choice is to interpolate the last melting only, since the last freezing is the one that leaves the final grains.

Size. In E1's slab about 60 cells of 0.5 mm have records. Each becomes 125,000 cubes of 10 micron, about 8 million cubes in all, or about 23 million records with the re-melting: roughly 1.1 GB in the binary format and 2 GB as text. A bead with full coverage is about 2.5 times larger. Keeping only the last melting cuts these figures by about three.

9.2 Recorder version 2 (proposed)

- **A tolerance band on the metal fraction.** A cell counts as metal once it is above 50% metal, and stops counting only when it falls below 10%. Ripples in between no longer throw records away (cause c).
- **'Arrived mushy' records.** When metal arrives in a cell between 815 K and 906 K, the recorder logs it with tm and tl both set to the arrival time and cr measured over the next 10 K (cause b). These records are flagged, so the export can write the ExaCA file both with and without them. The approximation to state in the thesis: ExaCA will start counting undercooling from zero on arrival, although the metal is really already up to 91 K below the liquidus and partly solid.
- **The peak temperature of every record**, for the smooth resampling.
- **A coverage report in the export**, so every dataset states how much of the bead it covers and how many records came from each route.

9.3 The runs that follow

dataset	what it is	good for
E1 (now)	M2 physics, 0.5 mm, recorder version 1; 19% coverage	setting up and testing the whole ExaCA chain
E2 (next, overnight)	the same run with recorder version 2	a much fuller dataset for trial ExaCA runs; still M2's narrow pool, so still not thesis results
calibrated 0.25 mm runs (remote PC)	heat source reshaped so the melted zone matches Zhao's width; cells half the size, so a droplet spans about 370 cells instead of 45 and G is sharper	the grain-structure results of the thesis; coverage becomes a calibration target next to width and depth
magnetic cases	the same pipeline with a field applied	the thesis question

10. The files to send to the ExaCA side

file	what it is	use it for
`exaca_waam-arc-01-E1-exaca.csv`	The ExaCA temperature file for the whole track: 3,610 records, columns x,y,z,tm,tl,cr in metres, seconds and K/s, on the 0.5 mm grid.	the first ExaCA test at CellSize 500 micron; the input for resampling
`exaca_E1_slab_x14-16mm.csv`	The same records cut to x = 14 to 16 mm, the middle of the steady window.	the first resampled test domain
`gr_waam-arc-01-E1-exaca.csv`	For every record: x,y,z,tl,cr,G,R (G in K/m, R = cr/G in m/s).	plots and G-R maps; ExaCA does not read it
`README.txt`	Definitions, units, grid, coordinates, material values and caveats, with the coverage and resolution warnings added on 20 September 2026.	read first
`E1-ExaCA-Data-Explained.pdf`	This note.	the background

Not needed: the OpenFOAM case itself, its processor folders and snapshots, and the recorder's raw per-core files, which the export has already merged and checked. The whole package is about 1.5 MB and fits in an e-mail.

Numbers the ExaCA side will need, all of which are in the README: liquidus 906 K, solidus 815 K, freezing range 91 K; plate top at z = 0; torch at x = -10 mm at t = 0, moving in +x at 20 mm/s; process 135 A, 16 V, wire 1.2 mm at 150 mm/s, single bead on a 4 mm plate (Zhao et al. 2021).

11. Glossary

term	meaning
alpha.metal	the fraction of a cell's volume filled with metal, 0 to 1 (volume of fluid method); the recorder counts a cell as metal above one half
bead	the ridge of solid metal left behind the moving torch
cell / cube	one small box of a simulation grid: 0.5 mm in the flow simulation, 2.5 to 10 micron in ExaCA
cellular automaton	a simulation in which each cell of a grid changes state by simple rules that look only at its neighbours

term	meaning
columnar / equiaxed	long, thin grains / grains of about equal size in every direction
cooling rate, cr	how many kelvin per second a point cools, measured here over the first 10 K below the liquidus
coverage	the share of the bead's cells that have at least one record; 19% in E1
dendrite	the tree-like branched shape in which a crystal grows; the spacing of its arms gets smaller when the metal cools faster
epitaxial growth	freezing that continues an existing grain and copies its lattice direction
G	the temperature gradient at the freezing front, K/mm or K/m
grain	one crystal of the metal, with its own lattice direction
interfacial response function	how fast a crystal tip grows at a given undercooling; ExaCA's material file
liquidus / solidus	906 K / 815 K for our Al-5Mg: fully liquid above the first, fully solid below the second
mushy zone	the range between solidus and liquidus, where solid and liquid coexist
nucleation	the appearance of a new grain in the liquid
R	the speed of the freezing front, R = cr ÷ G
record (event)	one melting and freezing of one cell: x, y, z, tm, tl, cr (and G)
resampling	turning values on a coarse grid into values on a finer grid
texture	a preference of the grains for one orientation
tm, tl	the time a location went above, then back below, the liquidus
undercooling	how far below the liquidus the liquid at a freezing front is
WAAM	wire arc additive manufacturing: building parts from beads of arc-melted wire

References

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ExaCA source code and input documentation: github.com/LLNL/ExaCA (examples/README.md; src/CAinputs.hpp for the HeatTransferCellSize notice, present from release 1.3.0 onwards).

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